Solution Requirements

Date	15 March 2026
Team ID	SWTID-2026-6686
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Team Leader & Members	**Leader:** MANISHA R (Size: 4) **Members:** GOPIKA K, VINDHIYA G, NITHIYASRI M
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	**Authentication**	*Role-based Single Sign-On (SSO) integration via Active Directory/ServiceNow ID to authenticate employees, line managers, and IT procurement agents securely.*	High
2	**Authorization levels**	*Role-based access matrix ensuring End-Users can only submit orders, Managers approve direct-report requests, and IT Admins access flow overrides & inventory tables.*	High
3	**External interfaces**	*Integration with ServiceNow Service Catalog, automated email notifications (SMTP), and API webhooks for hardware vendor fulfillment systems.*	Medium

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
4	**Transactions processing**	*Automated workflow execution via Flow Designer triggering laptop request tasks, approval routing, asset tagging, and catalog item status updates.*	High
5	**Reporting**	*Real-time IT dashboard generating fulfillment SLAs, order volumes, approval turnaround times, and pending laptop inventory metrics.*	Medium
6	**Business rules, etc.**	*Auto-approval logic for requests within budget thresholds (< $1,500) and automated escalation if line manager approval is pending > 48 hours.*	High
7	**Compliance to laws or regulations**	*Audit logging for all approval transactions, IT asset tracking compliance (ISO 27001), and alignment with company data retention rules.*	Medium
8	**Other**	*Automated notification triggers sending status updates to the requesting employee via email and portal notifications upon key milestone changes.*	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	**Performance & Speed**	*Flow Designer triggers and catalog rendering response speed under normal load conditions.*	*(e.g., Page loads in < 2 seconds, flow triggers in < 500ms)*
2	**Scalability**	*System capacity during peak onboarding periods across multiple geographic regions.*	*(e.g., Supports up to 5,000 concurrent laptop orders)*
3	**Security & Data Privacy**	*Data protection for employee personal information and hardware delivery addresses.*	*(e.g., End-to-end AES-256 encryption, TLS 1.3)*
4	**Reliability & Availability**	*Service availability for submitting requests and running automated procurement flows.*	*(e.g., 99.9% uptime per month)*
5	**Usability & Accessibility**	*User interface intuitive for employees requesting standard hardware without training.*	*(e.g., Complies with WCAG 2.1 AA standards)*
6	**Other**	*Maintainability of Flow Designer sub-flows and ease of updating catalog items.*	*(e.g., Modular flow architecture, 100% workflow logging)*